

Skel3D: Skeleton Guided Novel View Synthesis

Anonymous submission

Abstract

In this paper, we introduce a new approach for monocular open-set novel view synthesis (NVS) that leverages object skeletons to guide the underlying diffusion model. Motivated by applications in robotics, where abstract representations are crucial for planning object manipulation tasks, our method exploits the generality of skeletal structures to enhance view synthesis across diverse object types. Building upon a baseline that utilizes a pre-trained 2D novel view synthesizer model (Free3D), our method takes advantage of the Objaverse dataset, which includes animated objects with bone structures. By introducing a skeleton guide layer our approach provides additional structural information which can be exploited by the generative model and results in enhanced pose accuracy and multi-view consistency. In order to fairly evaluate our new approach, we propose a curated selection of Objaverse items, with a focus on those objects which possess well defined skeletons. We propose a simple but efficient mapping method from joint representations of animated 3D models to skeleton representations and generate a synthetic dataset of skeleton poses and views of the modelled object. Experimental results demonstrate that our skeleton-guided method significantly enhances consistency and accuracy across diverse object categories within the Objaverse dataset and out-performs existing state-of-the-art NVS techniques both quantitatively and qualitatively.

Introduction

Novel view synthesis (NVS) has emerged as a critical challenge in computer vision and graphics, aiming to generate new perspectives of objects or scenes from limited input views. Recent advancements, including Neural Radiance Fields (NeRF) (Mildenhall et al. 2021) and models based on diffusion methods (Sohl-Dickstein et al. 2015; Ho, Jain, and Abbeel 2020), have significantly improved the quality and efficiency of NVS. However, single-view NVS remains particularly challenging, as it requires inferring complex 3D structures from a single 2D image while maintaining structural consistency and pose accuracy across generated views. Current state-of-the-art approaches, such as Free3D (Zheng and Vedaldi 2024) and Zero-1-to-3 (Liu et al. 2023), have made substantial progress in single-view NVS by leveraging large-scale pre-trained diffusion models. These methods condition the generation process on camera poses and other implicit information. However, they can

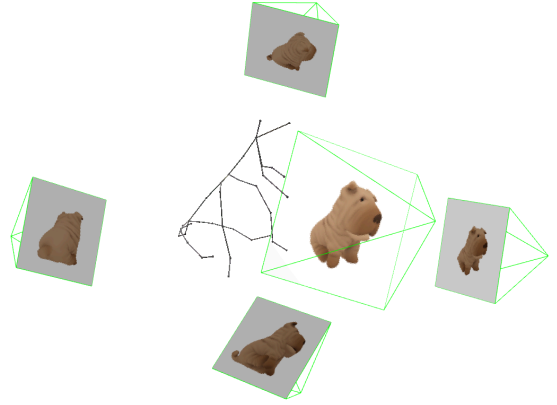


Figure 1: Using the predicted skeleton of the object as guide for novel view synthesis.

struggle with structural consistency and fine detail preservation, especially when dealing with complex geometries. The reliance on implicit information about object structure can lead to inconsistencies in generated views, particularly for out-of-distribution objects or unusual poses. In this paper, we introduce Skel3D, a novel approach to single-view NVS that leverages explicit structural information in the form of object skeletons. Our method is inspired by the success of skeleton-based techniques in related fields, such as human pose estimation and character animation (Kocabas et al. 2024). By incorporating skeletal information as a strong yet flexible prior, Skel3D aims to enhance both the structural consistency and pose accuracy of generated novel views. The key innovation of Skel3D lies in its use of a Skeleton Guide layer, which injects structural information directly into the diffusion process. Unlike existing methods that rely solely on camera pose information, our approach provides the model with explicit cues about the estimated pose and structure of the object. Crucially, we derive this skeletal information from a common spatial structure and project it into 2D for each desired view. As shown in Figure 1, this ensures that the underlying shape and structure of the object remains consistent across multiple generated views, addressing a significant limitation of current methods. To support the development and evaluation of Skel3D, we introduce

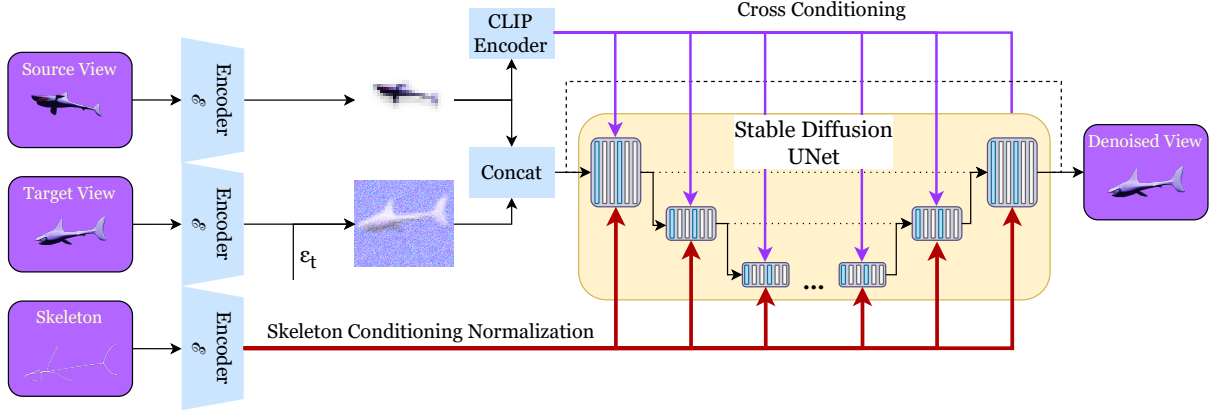


Figure 2: Architecture of our skeleton-guided model for NVS. Given a single input image, we introduce a Skeleton Conditioning Normalization (red) that utilizes the skeleton image embedding, enhancing the model’s capability to capture more precise views.

a curated dataset derived from Objaverse, focusing specifically on animated objects with bone structures. We carefully curated this dataset to include a diverse range of objects that are animated using skeletal systems, providing a rich source of data for training and evaluating skeleton-guided NVS models. This dataset not only enables the current work but also opens up possibilities for future research into temporally consistent object synthesis.

Our main contributions can be summarized as follows:

- We introduce the Skeleton Guide layer, a novel mechanism for incorporating skeletal information into the diffusion-based novel view synthesis process.
- We present a curated set of objects derived from Objaverse, specifically curated to include objects with skeletal animations, which aims to motivate further research in skeleton-guided NVS.
- We provide a comprehensive evaluation demonstrating that Skel3D outperforms existing state-of-the-art methods across multiple quantitative metrics and qualitative assessments.

Experimental results show that Skel3D consistently outperforms the current state-of-the-art baseline model according to each evaluated metric. Our approach demonstrates superior performance in maintaining structural consistency and pose accuracy, particularly for objects with well-defined skeletal structures.

The potential applications of Skel3D extend beyond static object rendering. By leveraging the temporal information inherent in skeletal animations, our approach paves the way for future work in space-time consistent object synthesis. This could enable more realistic and coherent animations from single-view inputs, with potential applications in fields such as computer graphics, augmented reality, and computer-aided design. In the following sections, we first review related work in novel view synthesis and skeleton-based modelling. We then provide a detailed description of

the Skel3D method, including the Skeleton Guide layer and our learning formulation. Next, we present our experimental setup, including the curated Objaverse dataset and evaluation metrics. We follow with a comprehensive analysis of our results, comparing Skel3D to existing methods and examining the relationship between skeleton quality and model performance. Finally, we discuss the limitations of our approach and potential directions for future work before concluding the paper.

Related Works

Our work on skeleton-guided novel view synthesis builds upon several areas of research in computer vision and graphics. In this section, we review the relevant literature and contextualize our contributions.

Novel View Synthesis (NVS)

Novel view synthesis has been a longstanding challenge in computer vision and graphics, aiming to generate new perspectives of objects or scenes from limited input views. Recent years have seen significant advancements in this field, particularly with the introduction of neural rendering techniques. Among these, Neural Radiance Fields (NeRF) (Mildenhall et al. 2021) has been particularly influential, demonstrating impressive results in synthesizing novel views of complex scenes.

However, the most relevant recent developments for our work are the landmark papers Free3D (Zheng and Vedaldi 2024) and Zero-1-to-3 (Liu et al. 2023). These works have pushed the boundaries of what is possible in single-view NVS by leveraging large-scale pre-trained diffusion models (Rombach et al. 2022; Lambdalabs 2022). Zero-1-to-3 introduced a framework for generating multiple views from a single input image using a diffusion-based approach. Building upon this, Free3D made significant improvements by introducing the ray conditioning normalization (RCN) layer. This innovation allowed for more efficient transfer of

target view information to the model, resulting in improved pose accuracy and multi-view consistency.

Single-View NVS and Generative Models

Single-view NVS presents unique challenges, as it requires inferring complex 3D structures from a single 2D image. Generative models, particularly diffusion models, have shown great promise in addressing these challenges (Tseng et al. 2023; Liu et al. 2023; Zheng and Vedaldi 2024; Liang et al. 2024; Voleti et al. 2024). The works of Free3D and Zero-1-to-3 demonstrate how these models can be conditioned on camera poses and other implicit information to generate novel views. However, current approaches still struggle with maintaining structural consistency and fine detail preservation, especially when dealing with complex geometries, deformations or out-of-distribution objects. The reliance on implicit information about object structure can lead to inconsistencies in generated views, particularly for unusual poses or complex, deformable objects.

Skeleton-based Modeling and Animation

Skeleton-based modeling has long been a fundamental technique in computer graphics, particularly for character animation (Akhter et al. 2008).

A skeleton is a hierarchical structure representing the underlying framework of an animated object, built upon a series of interconnected joint points. These joints serve as key pivot points for the object’s movement and transformation. The bones in the skeleton are the connections between these joints, defining the relationships and constraints of movement between different parts of the object.

This concept aligns with the principles of non-rigid structure-from-motion (NRSfM), where a non-rigid object is modeled as a linear combination of rigid structures. Similarly, in the utilized Objaverse (Deitke et al. 2022) dataset, the skeleton is designed to allow for complex, realistic animations by treating the animated object as a composite of these rigid components. The designers of Objaverse animations leverage this approach to create fluid and natural movements, adhering to the same foundational ideas that govern NRSfM.

In our work, the use of skeletons provides a powerful means of representing the underlying structure of objects. This approach has been widely used in areas such as human pose estimation (Cao et al. 2017) and character animation, demonstrating its effectiveness in capturing and manipulating object structure and pose.

Conditional Generative Models in Computer Vision

A key inspiration for our work comes from ControlNet (Zhang, Rao, and Agrawala 2023), which demonstrated the ability to control the pose of generated objects in synthesized images by incorporating external conditional information, such as a sketch or even skeletons. This work showed the potential of using structural guides to improve the output of generative models. The success of ControlNet in using skeletons to guide image generation led us to explore

whether similar structural information could enhance the performance of NVS tasks. Our approach extends this idea by integrating skeleton information directly into the NVS process through a novel skeleton normalization layer.

3D Reconstruction and Pose Estimation from Single Views

Recent advancements in 3D reconstruction from single images are also relevant to our work. Notably, the 3D-LFM (Lifting Foundation Model) (Dabhi, Jeni, and Lucey 2024) demonstrated that it’s possible to infer 3D skeletal structures from single images for a wide range of objects (Xu et al. 2022; Hirschorn and Avidan 2023). This aligns perfectly with the input scenario of single-view NVS and provides a potential method for obtaining the 3D skeletal information needed for our approach.

Bridging the Gap: Skeleton-Guided NVS

Our work, Skel3D, brings together these loosely related strands of research to address some of the present limitations of existing NVS methods. By incorporating explicit skeletal information into the diffusion process, we aim to improve both structural consistency and pose accuracy in generated novel views. Similar to how Free3D (Zheng and Vedaldi 2024) introduced the RCN layer to more efficiently transfer view information, we introduce a skeleton normalization layer that injects structural information about the object’s pose directly into the model. This approach combines the strengths of skeleton-based modelling with the generative power of diffusion models, potentially opening new avenues for high-quality, structurally consistent novel view synthesis. Furthermore, this approach opens up the way for generative models able to synthesize high-precision novel views for animated or deformed objects as well. In the following sections, we will detail our method and demonstrate how this skeleton-guided approach leads to significant improvements over existing state-of-the-art NVS techniques.

Method

Overview

Skel3D is a modern approach to single-view novel view synthesis that leverages skeletal information to guide the generation process. Our method builds upon recent advancements in diffusion-based image generation, introducing a Skeleton Guide Layer that incorporates explicit structural information into the synthesis process. This approach aims to improve both structural consistency and pose accuracy in generated novel views.

Skeleton Extraction and Representation

Rather than extracting skeletons from 2D images, we leverage the rich 3D information available in the Objaverse (Deitke et al. 2022) dataset, which contains a wide variety of objects with provided skeleton information. We prepared a curated dataset and multi-step rendering pipeline to generate the required data. We began with a high-quality subset of 12K objects curated by the Diffusion4D (Liang

et al. 2024) project, this is a manually filtered subset of Objaverse items suitable for animations. From this, we selected 260 objects as an isolated test set. Due to the utilisation of this original dataset for training the original Free3D backbone model which we extend, only objects not included in the original training data were selected for the test set. A full breakdown of the train/test object identifiers can be found in the supplementary material.

Each object was imported and rendered in Blender 4.2 (Blender Online Community 2024), preserving its original bone structure. The scene is prepared by resetting it, setting up the camera and lighting, and hiding mesh objects from rendering while keeping them visible in the viewport. We create geometric representations of the bones using icospheres at the bone heads and cylinders between adjacent bones. These representations are parented to their respective bones to ensure they follow the animation correctly.

For each object, we render every fourth frame of the first 24 animation frames, providing a diverse set of poses. All views were rendered with the background set to white, and render settings configured for ‘high-quality’ output. We save pairs of both the final view render and a special frame that includes the skeleton only, providing clear visualizations of the bone hierarchy and movements. This process results in a set of skeleton images that correspond to various poses of each object, providing rich structural information for our model.

Diffusion Model Architecture

We build upon the architecture used in previous works like Free3D and Zero-1-to-3, which leverage pre-trained latent diffusion models (Rombach et al. 2022). We utilize the same image-to-image Stable Diffusion checkpoint (Lambdalabs 2022) which consists of an image-to-image autoencoder (with encoder $\mathcal{E}$ and decoder $\mathcal{D}$) and a denoising diffusion model that works on the latents of this autoencoder. The diffusion model utilizes a UNet backbone that is mostly built from 2D convolution layers and cross-attentions as outlined in (Rombach et al. 2022). With the sole exception that the CLIP (Radford et al. 2021) embeddings used in these finetunes are from the image encoder instead of the text encoder.

We closely follow the training procedure and architecture of Free3D (Zheng and Vedaldi 2024) with the following notable changes:

- We do not use shared multi-view attention and multi-view noise sharing.
- We replace all Ray Conditioning Normalization layers with Skeleton Conditioning Normalization. (Except for the Skel3D+RCN, where we keep the RCN layer as well, see Table 1.)
- We use the image encoder $\mathcal{E}$ to create skeleton embeddings. Instead of using Fourier bands (in Free3D primarily motivated by the replaced ray information). We use the encoded features as conditions directly.

Among these, our key modification is the integration of the Skeleton Guide Layer into the existing UNet structure as visualized in Figure 2. By modulating the sub-modules

of the UNet with skeleton embeddings, we benefit from the inherent image generation capabilities of the pre-trained model while incorporating our structural guidance. This approach allows us to avoid retraining the entire network from scratch, focusing instead on fine-tuning the skeleton conditioning process.

Skeleton Guide Layer

The original form of Ray Conditioning Normalization (RCN) (Zheng and Vedaldi 2024) defines an adaptive layer normalization at each level of the UNet, where scale and shift parameters are produced from the ray condition vectors via an MLP. The key contribution of Skel3D is the Skeleton Guide Layer, using Skeleton Conditioning Normalization (SCN), which we implement by adapting the condition signal in the original RCN to utilise skeleton embeddings.

This layer combines adaptive layer normalization (Huang and Belongie 2017) (Dumoulin, Shlens, and Kudlur 2016) with skeleton conditioning to modulate the image latent. For each activation latent F_i of the i -th layer in the UNet, we apply the following steps:

1. Layer Normalization (LN)

$$LN(F_i) = \frac{F_i - \mu}{\sigma}, \quad (1)$$

where μ and σ are the mean and standard deviation of the activations F_i . In our use case following earlier practices (Zheng and Vedaldi 2024) we opt-in for a more restricted layer normalization variant, the Group Normalization (Wu and He 2018).

2. Skeleton Conditioning Normalization (SCN)

$$ModLN_{SCN}(F_i) = LN(F_i) \cdot (1 + \gamma) + \beta, \quad (2)$$

where γ and β are the scale and shift parameters predicted from the skeleton embeddings s via a multi-layer perceptron (MLP):

$$(\gamma, \beta) = MLP_{mod}(s). \quad (3)$$

This modulation is applied to each sub-module of the UNet, ensuring that the structural information provided by the skeleton effectively guides the image generation process.

Loss Function and Training Procedure

We maintain the original conditional DDPM-like loss function also used in previous works (Zero-1-to-3 and Free3D), as it has proven effective for novel view synthesis tasks. The learning objective is defined as

$$\mathcal{L} = E_{(\mathbf{z}_0^{\text{tgt}}, z^{\text{src}}, \mathbf{S}), \epsilon, t} \left[\epsilon - \epsilon_{\theta}(\mathbf{z}_t^{\text{tgt}}, t, \mathbf{S}, z^{\text{src}})^2 \right], \quad (4)$$

where $\mathbf{z}_0^{\text{tgt}} = \{\mathcal{E}(x_i)\}_{i=1}^N$ are the encoded target views, $z^{\text{src}} = \mathcal{E}(x^{\text{src}})$ is the encoded source view and $\mathbf{S} = \{\mathcal{E}(s_i)\}_{i=1}^N$ are the encoded target skeletons. A given training sample is then comprised of $(\mathbf{z}_0^{\text{tgt}}, z^{\text{src}}, \mathbf{S})$. The network is conditioned on the source view and skeletons for the target views and estimates ϵ noise values through the ϵ_{θ}



Figure 3: The first column shows the source image for NVS, followed by the target view in the second column. The third column presents the skeleton guidance used in the process. The fourth column, highlighted with green values, demonstrates the superior performance of our model. The final column shows the Free3D results, with red values indicating areas where our model outperforms.

estimator for all target views. There is no information flow between skeletons and target views of different skeletons.

We maintain the approach used in (Liu et al. 2023; Zheng and Vedaldi 2024), in which we concatenate the input image code z^{src} with each z_t^{tgt} along the channel dimension and use this as a latent input for the UNet. We leave the original Stable Diffusion base model implementations unchanged, and we keep the CLIP (Radford et al. 2021) encodings as cross-conditioning inputs for the cross-attention layers in the UNet.

Each model underwent a single training run utilising 8 A100 GPUs with 40GB of memory each. To align with our specific hardware setup and dataset size we adapt the training procedure as detailed below. Due to memory constraints,

we used a smaller batch size (32) compared to the original models. To compensate, we accumulated gradients over two steps, effectively increasing our batch size. Training was completed using the 12K high-quality subset curated by the Diffusion4D project (Liang et al. 2024). We fine-tuned the pre-trained diffusion model, focusing on integrating the Skeleton Guide Layer and optimizing its parameters. The training over 10 epochs took two days to complete.

Implementation Details

Our implementation builds directly upon the codebase of Free3D and Zero-1-to-3, with the primary addition being the Skeleton Guide Layer. We did not introduce any preprocessing steps or data augmentation techniques beyond the

skeleton rendering process described earlier. One challenge we faced was ensuring the quality and consistency of the skeleton renderings across a diverse range of objects. By using the curated subset from the Diffusion4D project (Liang et al. 2024), we mitigated some of these issues, but future work could explore more robust skeleton extraction and representation methods. In cases where the skeleton extraction might be inaccurate or incomplete, our model relies on the strength of the underlying diffusion model to generate plausible views. However, the quality of the skeleton information directly impacts the structural accuracy of the generated views, highlighting the importance of high-quality skeleton data.

Results

We evaluated Skel3D on a subset of the Objaverse dataset. Our evaluations were conducted using a single Nvidia A100 GPU with 40 GB of memory. In order to accurately assess the performance of the Skel3D method, we evaluated performance over the following metrics:

- **L1 Loss:** Measures the average absolute differences between predicted and ground truth images.
- **Structural Similarity Measure (SSIM):** Assesses the perceived quality of images.
- **Peak Signal-to-Noise Ratio (PSNR):** Evaluates the ratio between the maximum possible power of a signal and the power of corrupting noise.
- **Learned Perceptual Image Patch Similarity (LPIPS):** Quantifies the perceptual similarity between images.
- **Fréchet Inception Distance (FID-Score):** Measures the similarity between generated images and real images.

These metrics provide a comprehensive evaluation of both pixel-level accuracy and perceptual quality between the Free3D baseline and Skel3D.

Quantitative Results The performance of the Skel3D model across the targeted metrics are given in Table 1. In our experiments we observed that performing the additional fine-tuning on the curated training set with the original Free3D architecture resulted in a decrease in performance compared to the original pre-trained Free3D network. This may be due to over fitting of the original model as the selected training set objects were also in the original training set for Free3D. Therefore we compare directly to the original Free3D architecture with no additional fine-tuning. Across all metrics, we find the addition of the Skeleton Guide Layer leads to significant improvements in both pixel-level accuracy (L1 Loss, PSNR) and perceptual quality (SSIM, LPIPS, FID-Score).

In order to validate the statistical significance of the observed improvements, we perform a non-parametric Mann-Whitney U test, between both the Free3D baseline and the Skel3D implementation. We compare the performance of the original Free3D model with both the vanilla Skel3D and Skel3D with Ray Conditioning Normalisation. We find that both vanilla and RCN enhanced models, that for all metrics the observed improvements are significant with $p < .01$.

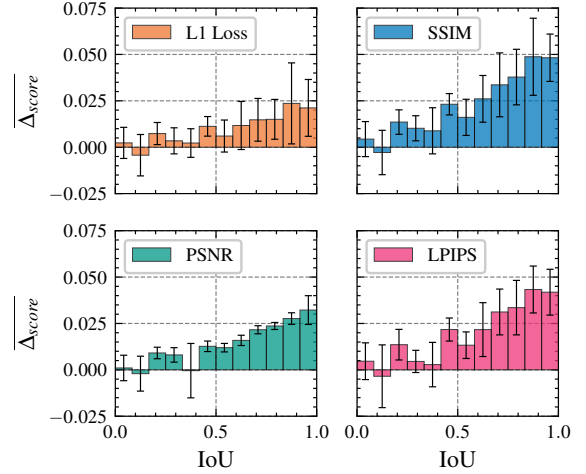


Figure 4: Average improvement in score depending on the quality of the skeleton. The x-axis represents the IoU of the bounding boxes of the object and the skeleton, which measures how well the skeleton fits the object. The y-axis shows the average improvement in metric scores, with error-bars given by the bootstrapped estimate of the standard error. Metrics where lower values are better (L1 Loss, LPIPS), were inverted by multiplying by -1 , and PSNR was scaled by a factor of 0.01 for better comparability.

While we observed in our evaluations that the combination of the skeleton guide layer and ray conditioning normalisation results in the best average metrics, we do not find the differences to be statistically significant. This suggests that skeleton guide layer is the dominant contributor to the performance improvement. A full listing of the statistical comparisons are given in the supplementary material.

Qualitative Results Figure 3 showcases examples where Skel3D significantly improves view generation compared to the baseline. These cases highlight how the incorporation of skeleton information leads to more accurate pose estimation and better preservation of object structure across different viewpoints. Conversely, Figure 5 presents examples where the baseline model performs better than Skel3D. Analysis of these cases reveals that the quality of the skeleton plays a crucial role in the performance of our method. When the skeleton poorly represents the object’s structure, it can lead to suboptimal results.

Figure 4 illustrates the correlation between skeleton quality and model improvement. The x-axis represents the Intersection over Union (IoU) of the bounding boxes of the object and its skeleton, serving as a measure of how well the skeleton fits the object. The y-axis shows the average improvement in metric scores. To ensure a positive correlation with performance improvements, we normalized and adjusted the metrics. L1 Loss and LPIPS were inverted by multiplying by -1 , and PSNR was scaled by a factor of 0.01 for better comparability. The plot demonstrates a clear trend, better-fitting skeletons (higher IoU) lead to significant improvements across all metrics.

Method	L1 Loss ↓	SSIM ↑	PSNR ↑	LPIS ↓	FID-Score ↓
Free3D (Zheng and Vedaldi)	0.0414 ± 0.0612	0.8711 ± 0.0999	19.848 ± 6.665	0.0935 ± 0.0931	2.6484
Skel3D	0.0335 ± 0.0503	0.8889 ± 0.0884	20.944 ± 6.432	0.0790 ± 0.0801	2.4697
Skel3D+RCN	0.0321 ± 0.0449	0.8926 ± 0.0841	21.124 ± 6.354	0.0747 ± 0.0746	2.4855

Table 1: Mean and standard deviation of the evaluated metrics for the original Free3D architecture as described in (Zheng and Vedaldi 2024), Skel3D without Ray Conditioning Normalisation (RCN) and Skel3D with RCN. Entries in bold indicate the best performance.

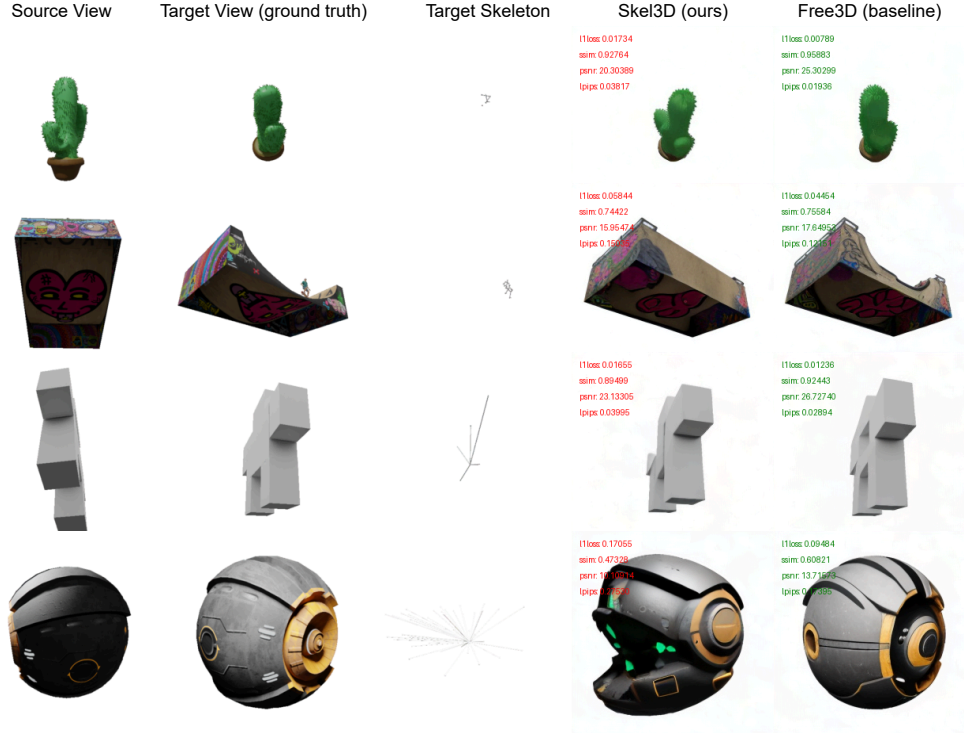


Figure 5: When the guidance skeleton is insufficient, our model’s performance drops compared to the baseline model. Best viewed online due to the small skeleton sizes compared to the object models.

Limitations and Failure Cases

Our analysis reveals that the performance of Skel3D is dependent on the quality of the skeleton information. Figure 5 demonstrates cases where the skeleton poorly represents the object’s structure or when skeleton information is insufficient, the performance can drop below that of the baseline model. This highlights the need for robust skeleton extraction methods to fully leverage the potential of our approach. Furthermore, we acknowledge that currently the current public datasets with readily available skeleton information are much smaller in scale than those typically used for NVS tasks. We highlight this as a potential further direction for extending the dataset. In-the-wild evaluations making use of 2D-3D skeleton lifting models (Dabhi, Jeni, and Lucey 2024) are also planned for future work.

Despite these limitations, Skel3D demonstrates good generalization across different object categories and pose types. The model’s ability to improve performance across a diverse

range of objects from the Objaverse dataset suggests that the skeleton-guided approach is effective for a wide variety of shapes and structures.

Conclusion

In conclusion, our results demonstrate that Skel3D significantly outperforms the state-of-the-art Free3D baseline when provided with high-quality skeleton information. The clear correlation between skeleton quality and performance improvement underscores the potential of skeleton-guided approaches for novel view synthesis, while also highlighting areas for future improvement in skeleton extraction and representation.

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Reproducibility Checklist

This paper

- Includes a conceptual outline and/or pseudocode description of AI methods introduced (yes)
- Clearly delineates statements that are opinions, hypothesis, and speculation from objective facts and results (yes)
- Provides well marked pedagogical references for less-familiar readers to gain background necessary to replicate the paper (yes)
- Does this paper make theoretical contributions? (no)
- Does this paper rely on one or more datasets? (yes) If yes, please complete the list below.
 - A motivation is given for why the experiments are conducted on the selected datasets (yes)
 - All novel datasets introduced in this paper are included in a data appendix. (partial)
 - All novel datasets introduced in this paper will be made publicly available upon publication of the paper with a license that allows free usage for research purposes. (NA)
 - All datasets drawn from the existing literature (potentially including authors' own previously published work) are accompanied by appropriate citations. (yes)
 - All datasets drawn from the existing literature (potentially including authors' own previously published work) are publicly available. (yes)

- All datasets that are not publicly available are described in detail, with explanation why publicly available alternatives are not scientifically satisfying. (NA)
- Does this paper include computational experiments? (yes) If yes, please complete the list below.
 - Any code required for pre-processing data is included in the appendix. (yes).
 - All source code required for conducting and analyzing the experiments is included in a code appendix. (yes)
 - All source code required for conducting and analyzing the experiments will be made publicly available upon publication of the paper with a license that allows free usage for research purposes. (yes)
 - All source code implementing new methods have comments detailing the implementation, with references to the paper where each step comes from (yes/partial/no)
 - If an algorithm depends on randomness, then the method used for setting seeds is described in a way sufficient to allow replication of results. (NA)
 - This paper specifies the computing infrastructure used for running experiments (hardware and software), including GPU/CPU models; amount of memory; operating system; names and versions of relevant software libraries and frameworks. (yes)
 - This paper formally describes evaluation metrics used and explains the motivation for choosing these metrics. (yes)
 - This paper states the number of algorithm runs used to compute each reported result. (yes)
 - Analysis of experiments goes beyond single-dimensional summaries of performance (e.g., average; median) to include measures of variation, confidence, or other distributional information. (yes)
 - The significance of any improvement or decrease in performance is judged using appropriate statistical tests (e.g., Wilcoxon signed-rank). (yes)
 - This paper lists all final (hyper-)parameters used for each model/algorithm in the paper's experiments. (yes/partial/no/NA)
 - This paper states the number and range of values tried per (hyper-) parameter during development of the paper, along with the criterion used for selecting the final parameter setting. (NA)